Sharzil Hasnine

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Portfolio: shas-9.github.io

Summary

Computer Science graduate (Adelaide University, Nov 2026) who builds working systems end to end: API-based data integrations, ETL pipelines, backend services and CI/CD. Strong in Python and SQL, comfortable reading, testing and correcting code, and experienced in turning stakeholder needs into delivered solutions that real users rely on.

Education

Bachelor of Computer Science (Advanced) Feb 2023 – Nov 2026
Adelaide University

Technical Skills

Data & Automation: Python, SQL (MySQL), REST APIs, ETL pipelines, data validation, Pandas, KNIME, REDCap
Engineering Practice: Git, code review, automated testing, CI/CD, Docker, Linux, Bash, Jira
Other Languages & Frameworks: TypeScript, JavaScript, Node.js, MongoDB, React, Java, Springboot, C++

Experience

Student Data Engineer - REDCap-KNIME Integration - Adelaide University Aug 2026 - Oct 2026
Tech: *Python, KNIME, REDCap, REST API*

- Connected a research data system (REDCap) to an analytics workflow tool (KNIME) by building a custom Python extension using PyCap to pull records and metadata through the REST API into separate typed output tables.
- Implemented metadata-driven ETL and validation: decoded categorical and checkbox fields, enforced field-level validation rules, and preserved longitudinal event and repeating-instrument context.
- Built a controlled test environment of 10,000 synthetic records across 64 fields (no real participant data), enabling reproducible validation of the extraction and ETL workflow.

Industry Project - Block Model Compression Algorithm - Maptek Aug 2025 - Oct 2025
Tech: *C++, Bash, Jira, CI/CD*

- Built a cross-platform CI pipeline with automated testing and reproducible Linux and Windows builds, integrating cross-compilation and artifact generation into the workflow and improving team velocity by 1.5x.
- Designed an automated, extensible testing framework that verifies compressed outputs against uncompressed input, independent of compression strategy.
- Architected an octree-based lossless compression system for 3D data achieving 98.8% size reduction and 2.6x faster processing than baseline, forming the foundation for team optimisations reaching 99.17% compression and 1520x speed.

Software Developer - Autonomous navigation - Adelaide Rover Team Oct 2024 - Feb 2025
Tech: *Python, C++, Ros2, Nav2, Gazebo*

- Designed a modular behaviour tree for the rover's autonomous navigation pipeline (Nav2) and validated behaviour in Gazebo simulation ahead of the Australian Rover Challenge.

Projects

Daily Iftar Registration System Feb 2026 - Mar 2026
Tech: *Typescript, Node.js, MongoDB, REST API, Docker*
Link: *aumsa.com.au*

- Delivered production backend features under tight release cycles for a live registration system used by 70+ daily users, working directly from user needs to deployed functionality.
- Built timezone-aware, DST-safe date handling (Australia/Adelaide), eliminating off-by-one scheduling bugs in day-based operations.
- Designed MongoDB schemas and API contracts that supported rapid feature expansion and clean frontend integration.

Paxos Consensus Algorithm Sep 2025 - Oct 2025
Tech: *Java, JUnit, Bash*
Link: *github.com/Shas-9/Paxos*

- Implemented Paxos from scratch with a TCP socket messaging protocol, and a failure simulation framework (latency, message loss, node crashes) to test correctness under partial system failure.

Distributed Weather Feed Aggregator with Lamport Clocks Aug 2025 - Sep 2025
Tech: *Java, JUnit, REST API*
Link: *github.com/Shas-9/Weather-Aggregation-Server*
- Built a multi-threaded data aggregation server with Lamport clock ordering, persistent storage with crash recovery, and automated expiry of stale data sources.

Leadership & Community

Secretary Nov 2024 – August 2026
ISSUA
- Led communications for a 10+ member committee, producing structured meeting minutes and action items to keep the team aligned and accountable.
- Managed university-facing logistics and administered the society's shared Google Drive records, keeping documentation organised and accessible.

Events Officer Nov 2023 – Nov 2024
ISSUA
- Planned and ran events for 100+ attendees, managing logistics, budget, venue and audio-visual setup, and liaising with stakeholders and guests.

Adelaide University Competitive Programming League Mar 2026 – Oct 2026
- Placed 2nd as a team in the 2026 league.